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Skill 22.1 Exercise 1

For each of the following indicate whether the example is an aggregate statistic.

Example	Aggregate Statistic (yes/no)
Median household income in a city	
A single student's test score	
A single household's income	
Minimum stock price over the last 30 days	

What is an aggregate statistic?

Skill 22.2 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following,

What is the most expensive shoe	
What is the least expensive shoe	
What is the number of different colors of shoes	
What is the average cost of shoes	
What are the different types of shoes	

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Skill 22.3 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following,

What is the most expensive shoe for each shoe type

What is the average cost of different colored shoes

What is the median cost for shoes of different material

What is the datatype returned for each statement above?

How can you convert each datatype into a dataframe?

Skill 22.4 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
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The following lambda function can be applied to return a value below which 25% of the data falls.

```
result = (lambda x: np.percentile(x, 25))
```

Use the lambda function above to answer the following,

What is the price threshold for the cheapest 25% of shoes of each type

What is the price threshold for the cheapest 10% of shoes of each material

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What is the price threshold for the cheapest 30% of shoes of each color	
What is the datatype returned for each statement above?	
How can you convert each datatype into a dataframe?	

Skill 22.5 Exercise 1

Refer to the *shoe_fly* dataframe below.

	id	first_name	last_name	email	shoe_type	shoe_material	shoe_color	price
0	41874	Kyle	Peck	KylePeck71@gmail.com	ballet flats	faux-leather	black	385.0
1	31349	Elizabeth	Velazquez	EVelazquez1971@gmail.com	boots	fabric	brown	388.0
2	43416	Keith	Saunders	KS4047@gmail.com	sandals	leather	navy	346.0
3	56054	Ryan	Sweeney	RyanSweeney14@outlook.com	sandals	fabric	brown	344.0
4	77402	Donna	Blankenship	DB3807@gmail.com	stilettos	fabric	brown	289.0

Write code to answer the following, store each result in a new dataframe.

What is the total number of shoes and each type and color?	
What is the average price of each shoe and color?	
What is the price threshold for the cheapest 30% of shoes of each color and type?	

Skill 22.6 Exercise 1

Refer to the *Skill 22.5 Exercise 1*

For each statement, use pandas to create pivot table.

What is the total number of shoes and each type and color?	
What is the average price of each shoe and color?	
What is the price threshold for the cheapest 30% of shoes of each color and type?	